ChatGPT

Portfolio Risk Management Framework

Executive Summary: For a total capital of ₹15,00,000 (with ₹5,00,000 active trading allocation), a disciplined risk framework is crucial. We recommend setting clear drawdown limits (daily/weekly/monthly), tight per-trade risk rules, diversified exposure caps, and structured rules for stops, pyramiding, and drawdown scenarios. Key elements include risk-per-trade (e.g. 0.5–1% of ₹15L), position-sizing by stop distance, maximum concentration limits (e.g. ≤10–15% per stock, ≤30–40% per sector), and total "open risk" across positions (ideally <3–5% of portfolio). Stop-loss orders (initial and trailing) should protect each trade, aiming for high reward-to-risk (≥2:1). In a losing streak (e.g. 3 or more losses), position sizes should be cut (the "step-down" approach) and trading halted after ~5 losses to review. As the portfolio grows (₹5L→₹15L), recalc risk sizes (e.g. ₹ risk per trade = 0.5–1% of updated capital) and rebalance exposures. We also define stress-test scenarios (sharp market crashes, volatility spikes, concentrated losses) to estimate worst-case drawdowns. A governance checklist (daily/weekly reviews, compliance with limits) and a structured monthly report (tracking P&L, drawdowns, rule breaches) will ensure discipline. The following sections detail each dimension with examples, tables of conservative/moderate/aggressive settings, and simulation guidelines.

1. Risk Limits (Drawdown Caps)

Define hard equity drawdown limits before stopping trading. A common rule is to cap drawdown at **10–15% of total capital** before a major review or halt. For example, a 10% max drawdown on ₹15L is ₹1.5L. Similarly, set **periodic loss limits** as percentages of total capital (not just the ₹5L trading segment):

- **Daily loss limit:** ~0.5–1% of ₹15L (₹7.5k–15k). Many traders stop trading if losses exceed ~2–3% in one day【40†L351-L359】.
- **Weekly loss limit:** ~1.5–3% (₹22.5k–45k), typically 2–3× the daily cap.
- **Monthly loss limit:** ~5–10% (₹75k–₹1.5L).

These limits ensure one bad day or week can't wipe out a significant chunk of capital【40†L351-L359】【56†L23-L31】. Hitting any stop-loss threshold should trigger an immediate trading pause and review. In other words, losses must be rare outliers, not routine【40†L351-L359】.

Conservative/Moderate/Aggressive Loss Limits: A useful way to illustrate risk appetite is via parameter tiers:

Parameter	Conservative	Moderate	Aggressive
Risk per trade (%)	0.5% (₹7.5k)	1.0% (₹15k)	1.5% (₹22.5k)
Daily loss limit (%)	0.5% (₹7.5k)	1.5% (₹22.5k)	2–3% (₹30k–45k)
Weekly loss limit (%)	1.5% (₹22.5k)	3% (₹45k)	6% (₹90k)
Monthly drawdown (%)	5% (₹75k)	7% (₹1.05L)	10% (₹1.5L)

Parameter	Conservative	Moderate	Aggressive
Max total drawdown (%)	10% (₹1.5L)	15% (₹2.25L)	20% (₹3L)

Sources: Professional traders often use ~1–2% risk per trade【6†L220-L224】【27†L186-L194】. Many "prop firm" style rules cap daily drawdown at ~4–6% (i.e. 2–3× per-trade risk)【56†L23-L31】【56†L75-L83】. These sample tiers can be adjusted for individual risk tolerance and horizon.

2. Per-Trade Risk & Position Sizing

Each trade's risk should be a small fixed percentage of total capital【27†L186-L194】【6†L220-L224】. Two common rules:

- **Conservative:** risk = 0.5% of ₹15L = ₹7,500 per trade.
- **Moderate:** risk = 1% = ₹15,000 per trade.

(This 0.5–1% range is widely recommended to survive losing streaks【52†L163-L172】【6†L220-L224】.) The position size (number of shares/units) is then calculated by dividing the risk amount by the distance to the stop-loss. For example:

- If risking ₹7,500 (0.5%) with a 2% stop loss, position ≈ ₹7,500 ÷ 0.02 = **₹3,75,000** notional (e.g. 3750 shares at ₹100 each).
- If stop is 1%, position ≈ ₹7,500 ÷ 0.01 = **₹7,50,000** notional.

Tighter stops allow larger positions (since you risk less per unit), provided the stop is technically justified. Conversely, a 1% risk (~₹15,000) with a 2% stop gives position ≈ ₹7,50,000.

Example Positions (0.5% vs 1% risk):

Risk per Trade	Stop-loss	Position Notional	Shares (at ₹100)
₹7,500 (0.5%)	1%	₹7,50,000	7,500
₹7,500 (0.5%)	2%	₹3,75,000	3,750
₹15,000 (1%)	1%	₹15,00,000	15,000
₹15,000 (1%)	2%	₹7,50,000	7,500

Note: The actual number of shares will be rounded to whole units and may be constrained by lot size or margin requirements (for derivatives). The key is that **risk in rupees is fixed in advance**. This "fixed fractional" sizing approach is the cornerstone of surviving drawdowns【25†L91-L100】【52†L163-L172】. Always set your stop-loss (at entry) first, then compute size so that a stop hit equals the pre-set risk【6†L220-L224】【52†L163-L172】. Never increase position because you "feel confident"; keep sizing mechanical【40†L370-L378】.

3. Exposure Limits (Diversification)

To avoid concentration risk, cap exposure per position, sector, or instrument type:

- **Single-stock limit:** e.g. 10–15% of total portfolio. In a ₹15L portfolio, 10% = ₹1.5L per stock. This prevents one equity's move from dominating the portfolio. (Mutual funds often limit individual stock holdings; similarly, a retail investor should curb concentration.)
- **Sector limit:** e.g. 25–40% of portfolio in one industry. For instance, avoid having ₹6L+ (40%) in a single sector like banking stocks at once. If there's a sector-wide event, a high sector exposure can be very damaging.
- **Instrument type limit:** e.g. ≤20–30% in derivatives or leveraged trades. If using F&O, don't put all equity capital into margin; treat F&O notional positions with caution. Ensure enough *cash or unleveraged assets* (20–30% in cash/mutual funds) for liquidity and margin calls.

These are guidelines; the exact caps depend on risk tolerance. Conservative portfolios may use lower caps (5% per stock, 20% per sector) while aggressive can tolerate higher. Regardless, **never put "all eggs in one basket"**【48†L269-L278】.

(No direct citation; these are prudent diversification rules aligned with typical portfolio design.)

4. Open-Risk Aggregation ("Portfolio Heat")

Monitor the **total at-risk capital across all open trades**. For example, if you have 5 trades each risking ₹7,500, total open risk = ₹37,500 (2.5% of ₹15L). A prudent practice is to cap aggregate open risk at ~2–5% of capital. This ensures no cluster of positions can wipe you out. Some professional desk guidelines call this "portfolio heat"【44†L7-L13】.

- **Rule of thumb:** Keep sum of individual trade risk <3–5% of portfolio. E.g. if using 0.5% per trade, no more than ~5–6 simultaneous trades (6×0.5%=3%).
- If correlation is high (e.g. multiple stocks in same sector), consider them as one risk cluster. Adjust the count accordingly.

If total open risk limit is hit, refuse new trades until some are closed or risk is reduced. This is an **explicit control to prevent overleveraging your own system**【44†L7-L13】.

(Sources on this concept are scarce in retail literature, but the idea is widely taught in trading risk management.)

5. Stop-Loss, Trailing Stops, and Exit Rules

Initial Stops: Every trade must have a stop-loss (hard exit) defined by chart support/resistance, volatility, or risk tolerance. Common methods:

- Use technical levels (e.g. a recent swing low/high, moving average, trendline) to set stops【48†L223-L232】.
- Use volatility-based stops (e.g. 1.5× ATR) to avoid normal noise triggering you out【48†L239-L247】. For example, one guideline is placing stops at least 1.5× the 14-day ATR away【48†L239-L247】.

- Avoid “hope stops” – define them before entry and never move them farther away for fear of loss 【40†L370-L378】.

Trailing Stops: As a trade becomes profitable, a trailing stop can lock in gains. A trailing stop “moves” with the market, preventing big winners from reversing into losses【22†L122-L130】. For example, once your trade is +2R in profit, you might move the stop to breakeven. Then you could trail the stop by a fixed %, ATR, or a lagging moving average to ride trends. Investopedia notes that a trailing stop is “best suited for trending markets” and helps “lock in profits while limiting losses”【22†L127-L135】.

Reward-to-Risk (R-multiples): Aim for trades where potential profit ≥ 2× risk. A target like 2:1 or higher ensures that winners outweigh losers over time【8†L213-L220】. For example, if risking ₹1, assume a 2R target (profit ₹2). Track trade outcomes in R-units (profit/risk) to evaluate your edge【46†L37-L42】. Van Tharp’s “R-multiple” concept standardizes trade results by risk【46†L37-L42】.

Pyramiding (Scaling In): Adding to winning positions can boost returns in strong trends, but must be controlled. A simple pyramiding rule: after the initial entry, add only if the trend proves strong (e.g. price moves further in your favor by a set amount). The added size should be smaller than the base (e.g. half-size) to limit risk【23†L131-L140】【53†L1-L4】. TradersPost notes that pyramiding “maximizes profit potential” but warns “risk management becomes paramount...because multiple correlated positions increase exposure”【23†L129-L138】【53†L1-L4】. If pyramiding, always widen stops on earlier lots or use an overall pyramid-wide stop, ensuring total potential loss remains within your allowed risk.

Exit Rules: Define clear profit-taking or exit criteria. Options include fixed targets (stop at 2R), trailing exits, or exiting on technical signals (e.g. break of a moving average). Don’t exit winners for too little profit; high-performers in your R-track record often let winners run longer. A disciplined exit plan (partially profit-taking vs trailing) helps maintain performance consistency【46†L65-L74】【46†L95-L103】.

6. Losing Streak Rules & Recovery Plan

Even a good system will have losing streaks. Pre-plan how to respond:

- **Stop after consecutive losses:** For example, if you incur **3 losing trades in a row**, reduce position size (e.g. cut risk per trade by 50%). If you hit **5 losses in a row**, halt trading for the day or longer and perform a review【52†L175-L183】. This “step-down” approach is explicitly recommended by trading educators【52†L175-L183】. For instance, trade with 1% risk normally, drop to 0.5% after 3 losses, and to 0.25% (or cease trading) after 5 losses【52†L175-L183】. Only resume full size after a string of calm, profitable trades.

- **Review and adjust:** After a losing streak, stop trading immediately and analyze. Did all trades violate stops? Did market conditions change? Check if risk rules were followed. Often, streaks arise from a breakdown in discipline or a market regime shift【52†L161-L170】【52†L171-L179】. A trade journal is invaluable here.

- **Avoid “revenge trading”:** Never increase size or break rules to recoup losses. Stick to fixed rules regardless of recent P&L【40†L370-L378】【52†L175-L183】.

- **Reset plan:** If losses were due to flawed signals or unusual volatility, refine strategy or wait for better conditions before ramping back up. Many traders “demo-trade” or take a break after a drawdown, then re-enter gradually.

This disciplined approach ensures a drawdown does not spiral out of control. As one guide notes, proper risk management means even 10 consecutive losses only shrink 1–2% risk-per-trade systems by ~10–20%, which is recoverable【52†L163-L172】.

7. Rebalancing & Scaling with Growth

As your portfolio grows from ₹5L to ₹15L, recalc risks and reallocate gradually:

- **Scale risk per trade:** If you keep risk % constant, your rupee risk per trade rises (e.g. 0.5% of 10L = ₹5k, vs 0.5% of 15L = ₹7.5k). Recompute your position-size formula at each milestone. This way your absolute gains (and allowed losses) scale with capital, but the percentage risk stays fixed.

- **Rebalance portfolio:** Periodically (e.g. quarterly), rebalance to maintain target exposures. If certain stocks have run up, trim them if they exceed exposure caps. Shift gains into cash or new ideas.

- **Risk budget:** Consider adding higher-risk strategies only as capital grows. For example, if ₹5L was your learning/trial phase, move gradually to full-allocation at ₹15L. Avoid jumping leverage proportionally; let increasing cash cushion compound with the portfolio.

- **Adjust stop distances:** Larger capital can allow wider stops without increasing % risk, since position size adjusts. But always verify stops make sense technically, not just monetary.

- **Update mental thresholds:** As absolute amounts change, remind yourself of the new daily/weekly loss limits (e.g. ₹45k/month was 3% of 15L).

Assumptions: We assume all ₹15L is liquid. If ₹10L is tied up elsewhere (long-term), you might treat the ₹5L trading as a standalone account and risk accordingly.

8. Stress-Test Scenarios and Drawdowns

Run hypothetical scenarios to see potential drawdowns:

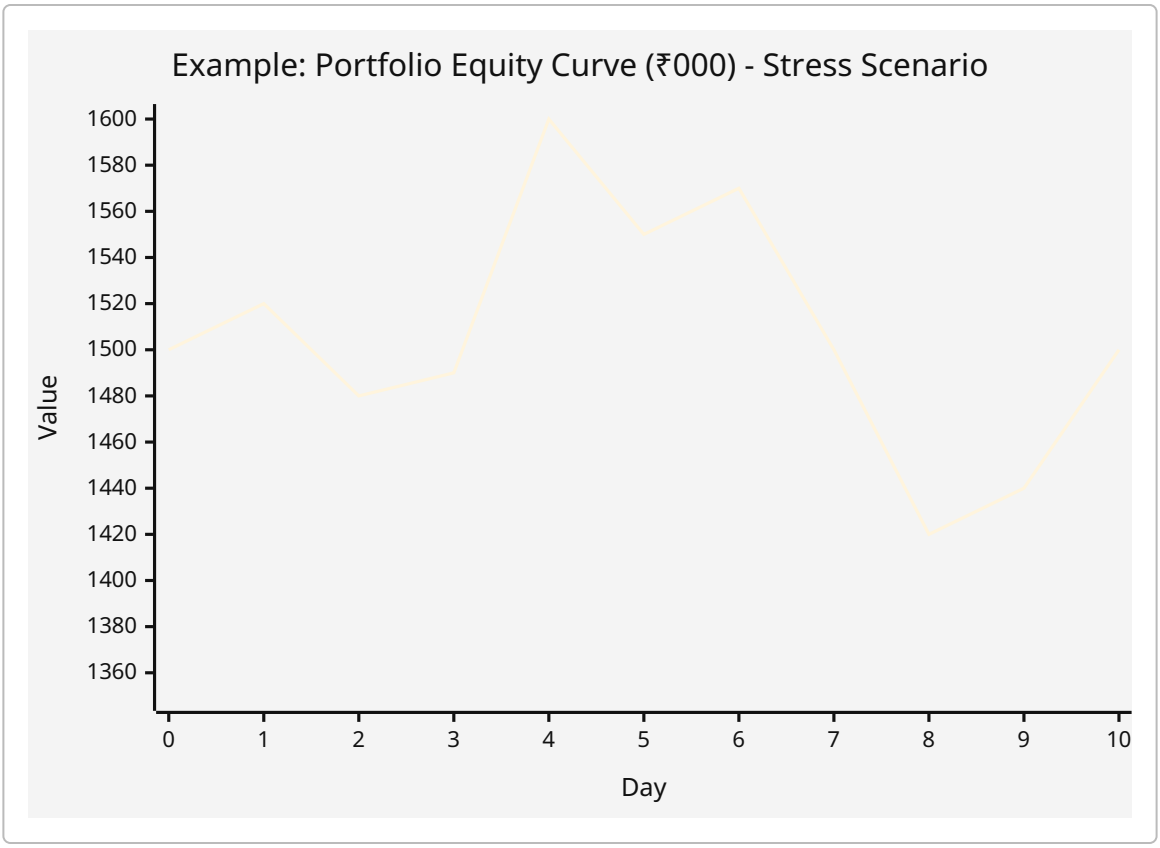
1. **Market Crash (Systemic):** Suppose a sudden equity crash (e.g. –25% in a week). If fully invested (evenly across 5 stocks with 2% stop on each, which might not help in a gap crash), worst-case might be ~25%×exposure. If we had 60% in stocks, crash = –15% of portfolio (₹2.25L loss). With stops, you might cut off partial losses earlier, but gap risk remains.
2. **Volatility Spike:** High volatility can trigger stops. For example, a 2×normal volatility day might stop out positions at 1.5× ATR. If you used volatility-based stops, you’d automatically take a big loss (e.g. a quick –3R on a large position = –% of capital). Simulate: if a 10% average move hits your 5% stop on a large position, could be –3% portfolio in one day.
3. **Concentrated Loss:** If a single position (10% of port) loses 15%, that’s a –1.5% hit. If it’s on margin (e.g. futures double leverage), risk could double.

A simple **drawdown table**: assume a conservative 5-stock equal-weight, 10% each, rest in cash for crash:

Scenario	Assumptions	Portfolio Loss
Normal 5% Dip	Moderate volatility, tight stops (1% risk)	~5% (₹75k)
Crash (–25%)	Sudden 25% equity drop; no time for stops	–15% (₹2.25L)
Vol Spike +10%	One-day swing triggers all stops (2% risk each)	~–10% (₹1.5L)
Single-stock dump	Largest position (10%) down 20% vs. no hedge	–2% (₹30k)

(These are illustrative. Real outcomes depend on exact stop placement and diversification.)

To visualize, a rough **simulated equity curve** under turbulent conditions might look like (chart is illustrative):



This hypothetical run shows a deep drawdown around Day 8 after two sharp losses, with partial recovery thereafter.

Insight: Deep drawdowns are hard to fully recover from. Preventing them (via stops and diversification) is better than “bouncing back”.

9. Monitoring, Governance & Review

Implement a systematic review process:

- **Daily Checklist:** Before market open, note account equity and set absolute loss triggers【56†L49-L57】. After market close, review each trade: Were stops honored? Did any break risk rules? Record P&L, largest move against you, and mark if any hard limit (daily loss) was hit. Check remaining open risk.
- **Weekly/Monthly Reports:** Consolidate metrics: starting vs ending portfolio value, net P&L (%), maximum drawdown that period, win rate, etc. Compare performance against benchmarks (or your own targets). A sample monthly report table might include:

Month	Start ₹	Inflow/ Outflow	End ₹	Net P/L (%)	Max Drawdown (%)	Notes/Actions
Jan 2026	5,00,000	+0	5,50,000	+10%	4%	Tightened stops on ABC
Feb	5,50,000	+0	5,40,000	–2%	3%	Noted sector concentration
Mar	5,40,000	+2,60,000 (added)	8,00,000	+40%	5%	Rebalanced portfolio
…	…	…	…	…	…	…

(This is a template; actual values vary. Include any breaches of risk rules and remedial actions.)

- **Review Meetings:** Set a regular cadence (e.g. weekly 30-min, monthly 1-hr) to go over results with a mentor or peer. Ensure the rules are enforced and not ignored after a winning spree.

- **Governance:** Document all risk rules in a written trading plan. Treat it like company policy – any rule breaches must be logged and explained. This creates accountability.

- **Compliance:** Maintain an electronic (or paper) **trading journal**. Record each trade's entry/exit/stop and outcome, plus notes. This makes the review factual (as recommended by risk literature【48†L269-L278】) and reveals if patterns (like moving stops or overtrading) occur.

(Sources: Inspired by professional trading protocols【56†L23-L31】【48†L269-L278】 and prop-trading rules; plus general best practices.)

10. Implementation Checklist and Reporting

Implementation Steps: As you put this plan into practice, follow an action checklist:

- [ ] **Define Rules:** Write down all risk limits and rules (risk per trade, stop placement method, drawdown caps, position/exposure limits, losing-streak plan).

- [ ] **Trading Plan Entry:** Integrate stops and targets into every trade setup. Use trading software alerts if possible.
- [ ] **Position Sizing Setup:** Use spreadsheets or tools to calculate position size from risk. (E.g. spreadsheet: RiskAmount = Account × Risk% ; Size = RiskAmount ÷ (Stop%)).
- [ ] **Journal Start:** Begin logging all trades with entry, stop, and exit in a journal. Ensure it timestamps and categorizes them.
- [ ] **Pre-market Prep:** Each day, review economic calendar and set per-day loss limits (e.g. via platform alerts)【56†L49-L57】【56†L75-L83】.
- [ ] **Periodic Review:** Schedule weekly/monthly reviews. Prepare the report table above with actual data.
- [ ] **Stress Test Bi-annually:** At least twice a year, re-run drawdown scenarios to see if risk rules need adjustment (especially as portfolio scales).
- [ ] **Compliance Audit:** Periodically (quarterly), check if any risk limits were violated (even if no loss resulted) and refine the plan.

Sample Monthly Report Template: Below is a simplified table for tracking performance and compliance each month:

Month	Start ₹	Contributions	End ₹	Net Gain/ Loss	% Return	Max Drawdown	% Drawdown	Rule Breaches / Notes
Jan 2026	500,000	0	550,000	+50,000	+10.0%	–20,000	–4.0%	None
Feb 2026	550,000	0	530,000	–20,000	–3.6%	–30,000	–5.5%	Daily loss limit hit (2%)
Mar 2026	530,000	70,000	660,000	+60,000	+11.3%	–15,000	–2.3%	Added capital; adjusted exposure
…	…	…	…	…	…	…	…	…

(Include any capital injections or withdrawals. The "Rule Breaches" column documents if, say, the daily loss cap was reached or a stop was missed.)

Governance: Anchor all of the above in a written document or digital binder. Treat your portfolio management like a mini-fund: decisions and deviations are recorded. This prevents rationalizing away mistakes and enforces the discipline that ensures long-term survival.

Sources: Risk guidelines are drawn from trading education and industry practices. For example, trading educators emphasize small per-trade risk (≈1%) and strict loss limits【27†L186-L194】【52†L163-L172】. Trailing stops and 2:1 reward rules are standard advice【8†L213-L220】【22†L127-L135】. Tax rates for equities (STCG 20%, LTCG 12.5% over ₹1.25L【20†L93-L101】【20†L102-L109】) have been updated as per India's Union Budget 2024. In all cases, we stress clear, quantitative rules and regular review – the bedrock of professional risk management.
